GOBIERNO DE CHILE · MINISTERIO DE ECONOMÍA, FOMENTO Y TURISMO

Executive Summary to the Government of Chile

Phase 2: Where the Fund and the Summer Visitors Should Go

Prescriptive decision models for the 2027 Regional Tourism Development Fund (Mandate A) and High Season 2027 (Mandate B), with the questions that connect them (Mandate C).

To: Office of the Vice Minister of Tourism, Ministry of Economy, Development and Tourism

From: Prolific Consulting (Group 1)

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Course: TGM 530 Big Data in the Global Economy

Models: Model A, 75 decision variables and 71 constraints; Model B, 90 decision variables and 37 constraints; both solved to global optimum in LINGO 22.0

Main body: pages 2 to 6. Appendices A to E follow: A formulation of both models in words and in math; B complete LINGO code for both base models; C LINGO Solution and Range reports for both base models; D scenario log; E generative AI log. All model files and reports are in the accompanying zip.

Our recommendation. Approve the model's USD 45 million Fund plan with one change, moving USD 1 million of Magallanes' allocation into accommodation INFRA backed by a Magallanes project office, and run High Season 2027 on our routing plan, buying Punta Arenas seats and Patagonian beds first.

The Fund delivers **USD 121.5 million** of incremental visitor spending in 2027 to 2029 (USD 2.70 for every dollar) and exactly the **1,600** promised jobs. The summer plan nets **USD 50.2 million** from **110,000** program visitors, fills every seat at all six gateways and sends 75% of visitors beyond Santiago and Valparaíso. The Magallanes change gives up USD 0.15 million of modelled spending (USD 0.09 million once the project office exists) and returns about USD 1.29 million over the 2028 to 2030 summers. **In Patagonia, beds and seats limit Chile.**

Who we are and why you can trust this work. Prolific Consulting delivered the RfP 1 diagnostic and the RfP 2 outlook that won this engagement. In Phase 1 we flagged the Ministry's data gaps, put two regions in Wait even when it was uncomfortable, and reported openly where our own forecasts lost to a naive seasonal rule. Phase 2 meets the same standard. Both models implement the Terms of Reference exactly (75 and 90 decisions, 71 and 37 rules), both reached the global optimum in LINGO, and an independent second solver reproduced every number in the two base LINGO reports. We report the cost of every rule together with the range over which that cost holds, and we re-solved every scenario instead of extrapolating a dual price beyond its range (Appendices C and D). Where the model and our own judgment disagree, we say so and tell you who we think is right.

1. The Fund: Model A (A-1)

Region	RfP 1 tier	TALENT	INFRA	PROMO	SMART	NATURE	Total	Ceiling	Binds at	Our RfP 2
Arica y Parinacota	Wait	0.90	–	–	–	0.30	**1.20**	2.5	floor	–
Tarapacá	Wait	0.34	0.86	–	–	–	**1.20**	2.5	floor	–
Antofagasta	Prepare first	–	2.40	2.10	–	–	**4.50**	4.5	ceiling	1.50
Atacama	Prepare first	–	1.54	1.46	–	–	**3.00**	3.0	ceiling	1.50
Coquimbo	Scale now	2.11	–	1.88	0.01	–	**4.00**	4.0	ceiling	5.25
Valparaíso	Scale now	2.19	–	1.94	–	–	**4.13**	5.0		6.75
Metropolitana	Scale now	–	–	–	1.79	–	**1.79**	4.0	SMART cap	6.50
O'Higgins	Prepare first	–	1.20	–	–	–	**1.20**	2.5	floor	1.00
Maule	Prepare first	1.83	–	1.67	–	–	**3.50**	3.5	ceiling	1.50
Biobío	Prepare first	–	–	–	1.20	–	**1.20**	4.0	floor	2.00
Araucanía	Prepare first	4.00	–	–	–	–	**4.00**	4.0	ceiling	3.00
Los Ríos	Prepare first	2.78	–	–	–	–	**2.78**	3.0		1.00
Los Lagos	Scale now	5.00	–	–	–	–	**5.00**	5.0	ceiling	6.25
Aysén	Prepare first	–	3.00	–	–	–	**3.00**	3.0	ceiling	1.75
Magallanes	Scale now	–	–	0.30	–	4.20	**4.50**	4.5	ceiling	5.50
Total		**19.15**	**9.00**	**9.35**	**3.00**	**4.50**	**45.00**			**43.50***

USD million, LINGO ModelA_Base (objective 121.5194). *Our RfP 2 total excluded the USD 1.5 million cross-cutting monitoring pilot. Patagonia shaded.

What the plan does. The Fund and the jobs commitment both bind: every dollar is spent and the plan creates exactly 1,600 jobs. Eight regions reach the most their governments can execute, four sit on the USD 1.2 million floor (Arica y Parinacota, Tarapacá, O'Higgins, Biobío), and Metropolitana is held by its own SMART cap. INFRA, SMART and NATURE sit exactly at their national minimums, while TALENT (19.15, or 43% of the Fund) and PROMO (9.35 of the 11.25 cap) have room. TALENT dominates for a clear reason: it creates the most jobs per dollar in every region, so the jobs rule pulls money toward it. Foundations first binds in six regions (Antofagasta, Atacama, Coquimbo, Valparaíso, Maule, Magallanes): there, every extra dollar of promotion needs USD 1.33 of training or infrastructure first, exactly the behaviour the rule was written to enforce. The macrozones all hold with room: Norte 13.9 (minimum 9.0), Sur 12.98 (minimum 11.25), Austral 7.5 (minimum 4.5) and Centro 10.62 (maximum 15.75), so balance costs the country nothing today.

Against our Part 1 judgment. The model agrees with us on Magallanes, Los Lagos and Coquimbo, all Scale Now regions that receive their full ceiling. It goes further than we did on our Prepare First regions, sending ceiling amounts to Aysén, Antofagasta, Araucanía, Maule and Atacama (USD 18.0 million against our 9.25 million), because their multipliers are high and their foundations programs earn well. It confirms our Wait tier: Arica y Parinacota and Tarapacá receive only the floor, and none of it goes to promotion. It disagrees sharply on Metropolitana (1.79 against our 6.5 million), whose INFRA and NATURE multipliers are the lowest in Chile. Our RfP 2 split would also break 11 of Model A's rules (it was written before they existed) and scored with the Ministry's multipliers it would deliver only USD 103 to 107 million, against 121.5. **Who is right?** The model is right on marginal returns, and we accept its plan. We were right that Santiago anchors corridor products and the national data platforms that the multipliers do not price, so Santiago's share goes to SMART, where it earns the most, and private co-funding should carry its promotion. **Recommendation:** approve the allocation as shown, disburse first to the eight regions at their ceiling, where execution capacity is the scarce resource, and attach the one Magallanes change in Section 5.

2. High Season 2027: Model B (B-1)

Gateway \ region	ARI	TAR	ANT	ATA	COQ	VAL	RM	OHI	MAU	BIO	ARA	LRI	LLA	AYS	MAG	Seats used
Santiago (SCL)	–	–	–	15.5	9.2	–	4.5	1.0	6.3	13.5	1.5	–	0.9	–	2.6	**55 of 55**
Northern airports	4.9	0.9	6.2	–	–	–	–	–	–	–	–	–	–	–	–	**12 of 12**
Puerto Montt (PMC)	–	–	–	–	–	–	–	–	–	–	–	1.1	8.9	2.0	–	**12 of 12**
Punta Arenas (PUQ)	–	–	–	–	–	–	–	–	–	–	–	–	–	–	8.0	**8 of 8**
Los Libertadores (land)	–	–	–	–	–	11.5	2.5	–	–	–	–	–	–	–	–	**14 of 14**
Cruise terminals	–	–	–	–	–	9.0	–	–	–	–	–	–	–	–	–	**9 of 9**
Visitors (thousand)	**4.9**	**0.9**	**6.2**	**15.5**	**9.2**	**20.5**	**7.0**	**1.0**	**6.3**	**13.5**	**1.5**	**1.1**	**9.8**	**2.0**	**10.6**	**110**
Binds at		min	beds	beds	beds		beds	min	beds	beds	min	min		beds	beds	

Thousand program visitors, January to March 2027, LINGO ModelB_Base (objective USD 50,246 thousand). "beds" = accommodation headroom binds; "min" = only the committed minimum.

What the plan does. All six gateways are 100% full, so seats, not demand, cap the program: one more seat is worth USD 384 to 484 depending on the gateway, far more than one more bed in most regions. Santiago feeds nine regions; the northern airports serve the three northern regions; Puerto Montt feeds Los Lagos, Los Ríos and all of Aysén; Punta Arenas sends its 8,000 seats to Magallanes, and 2,600 more Magallanes visitors still connect through Santiago. Eight regions reach their bed limit; four (Tarapacá, O'Higgins, Araucanía, Los Ríos) receive only their committed minimum, which costs USD 40 to 50 per visitor, about USD 200,000 a season. Decentralization binds at exactly 75%, at USD 15 per visitor moved.

Two people, two summers. The Aysén hotelier has a full house: 2,000 program guests, all via Puerto Montt, spending about USD 1.7 million, with not one spare bed. Her constraint is rooms, and more promotion would only turn guests away. **The Araucanía tour operator** receives only the 1,500 promised visitors while 14,000 spare beds stay empty, because a visitor there is worth USD 410 against 850 in Aysén. His season depends on raising spending per visitor (longer stays, paid experiences), not on routing. **Recommendation:** run the summer on this plan, keep the regional commitments (their USD 200,000 cost buys political consent cheaply), and spend negotiating effort on seats into Patagonia, not on promotion.

3. Stress testing the plans (A-2 to A-5, B-2 to B-7)

Q	Change	Dual predicts	Re-solve	What it means and what we advise
A-2	Fund 45 to 50 / 40	+15.83 / −15.83	+11.84 / −17.40	The Fund's dual price is 3.165: one more million return about USD 3.17 million, but only between USD 44.88 and 45.31 million (Range report). Five more million are worth less than 5 × 3.165 because each goes to a worse use: the PROMO cap and four more ceilings bind and the marginal value falls to 2.10. Five fewer costs more because the cuts fall on the best uses; the marginal value rises to 4.16. For the Budget Office: the first million cut costs about 3.2 million in visitor spending, the fifth about 4.2. **Defend the 45 million**; accept a top up only with higher ceilings.
A-3	Jobs 1,600 to 1,800	−3.67	−5.11	The jobs dual (−0.0184) means each extra job costs about USD 18,400 of spending, but only up to 1,604 jobs. Reaching 1,800 costs USD 5.11 million, or USD 25,600 per job, 39% more than the margin suggests, because the plan keeps shifting money from high return PROMO and NATURE into TALENT. **Advise against** unless the Presidency values a three-year tourism job above about USD 25,600 of visitor spending.
A-4	Floor 1.2 to 2.0	−0.64	−0.87	Today the floor costs Tarapacá 0.34, O'Higgins 0.26, Biobío 0.13 and Arica 0.06 per million. At 2.0, seven floors bind instead of four: Arica, Tarapacá, Metropolitana, O'Higgins and Biobío gain USD 3.4 million, paid for by Valparaíso, Los Ríos and Atacama, at a cost of 0.7% of total spending. *For:* every governor gets a program large enough to matter, and the Wait regions can build foundations (their extra money goes to TALENT, not PROMO). *Against:* USD 0.87 million of spending is forgone and money leaves higher return regions. **Support 2.0**, earmarked to TALENT or INFRA; re-solve before going higher.
A-5	SMART minimum 3 to 5	−2.08	−2.17	SMART is the costliest national rule (−1.04 per USD 1 million, against NATURE −0.34 and INFRA −0.28); the TALENT minimum and the PROMO cap cost nothing because they do not bind. At 5.0 the extra SMART goes to Coquimbo (to its 1.59 cap) and Tarapacá. The multipliers miss SMART's spillovers: better forecasts, crowd and park monitoring, and data for every other program. **Support only if** those benefits are credibly worth more than about USD 2.2 million; otherwise phase the increase into 2028.
B-2	PUQ seats +2,000 / +4,000	+967.5 / +1,935	+967.5 / +1,781	Punta Arenas has the highest seat dual (USD 483.75 per visitor), valid for 2,600 extra seats, so the +2,000 prediction is exact. A Punta Arenas seat is worth USD 100 more than a Santiago seat because it saves the Santiago connection (cost 70 against 170). Beyond 2,600 seats, Magallanes beds are full and extra seats only replace Puerto Montt flights to Aysén, so a seat is worth USD 373.75. **Negotiate Punta Arenas first, for about 2,600 seats.**
B-3	Value of 1,000 more beds	–	–	Ranking: Magallanes USD 430,000, Aysén 345,000, Metropolitana and Maule 60,000, Biobío 35,000, Atacama 20,000, Antofagasta 15,000, Coquimbo 10,000; the other seven regions zero. Two surprises: Santiago, with the most hotels, ranks only third, because its visitors spend little more than elsewhere; and Los Lagos is worth zero despite being a top destination, because it has 27,700 spare beds and its limit is seats. **Invest in Patagonian beds first.**
B-4	PMC seats 12,000 to 6,000	−2,452.5	−2,452.5	Exact, because the cut stays inside the range (down to 3,100 seats). The plan protects Aysén, which has no other route, and lets Los Lagos absorb the loss: Los Lagos falls from 9,800 to 5,300 visitors (Santiago backfills 1,500 of them) and Valparaíso loses 1,500. **Prepare a Santiago to Balmaceda contingency** before the season.
B-5	Decentralization 75% to 80%	≈ −82.5	−92.5	Cheap: 0.18% of program value. Valparaíso loses 5,500 visitors and Los Lagos gains 5,500 (it has the spare beds). **Support**, with crowding safeguards at Chiloé, where DB4 sensors already show trail pressure.

Q	Change	Dual predicts	Re-solve	What it means and what we advise
B-6	Magallanes spend 980 to 833	−1,558.2	−1,558.2	The routing does not change; only the value falls (10,600 visitors × USD 147). It holds until spending falls below about USD 550 (980 minus the USD 430 bed value). **The plan is robust to the peso**; seats and beds can be committed without waiting for the exchange rate.
B-7	Unused routes	–	–	Reduced costs: Santiago to Aysén needs a cost cut of more than USD 35 per visitor (160 to below 125); Puerto Montt to Magallanes more than USD 5 (150 to below 145). Levers: a program fare or airport fee waiver into Balmaceda, and a Puerto Montt to Punta Arenas codeshare bundle. **Negotiate both**: they are cheap insurance against a B-4 closure.

Model A in USD million of 2027 to 2029 spending; Model B in USD thousands of net contributions per season. A dual price holds only inside its Range report interval; outside it, we re-solved.

4. The governors (A-6) and execution capacity (A-7)

A-6. Neither request is in the optimal plan. The reduced costs are 0.1847 for ALLOC (METROPOLITANA, PROMO) and 0.1324 for ALLOC (MAGALLANES, INFRA): each multiplier would have to improve by that much before the model funded it, Santiago promotion from 2.65 to about 2.83 and Magallanes infrastructure from 3.69 to about 3.82. Forcing a full million into Santiago promotion costs more than the reduced cost suggests (USD 0.59 million in the LINGO re-solve), because foundations first would also require about USD 0.93 million of training there.

> *To the Governor of Metropolitana: "Santiago promotion returns USD 2.65 per dollar and would need to reach about 2.83 before it beat the alternatives elsewhere in Chile; a full USD 1 million moved into it would cost the country about USD 0.6 million in visitor spending. Your region's USD 1.79 million goes to smart destination platforms, where it earns the most and where Santiago leads the country."*
>
> *To the Governor of Magallanes: "Your region already receives its full USD 4.5 million, and inside that ceiling nature and promotion out-earn infrastructure, which would need to reach USD 3.82 per dollar to enter the plan on its own. Because our summer model shows your beds are Chile's most valuable bottleneck, we are recommending USD 1 million for accommodation anyway (C-1)."*

A-7. The two outlier absorption duals are Magallanes 1.528 and Aysén 1.468, about six times the next region (Antofagasta 0.255). Patagonia is limited by its regional governments' capacity to execute, not by money or ideas: each extra million they could spend would earn about one and a half million more than the same money elsewhere. A project office raising Magallanes' ceiling from 4.5 to 5.5 lifts the objective to 122.9943 (+USD 1.47 million); the dual holds for only the first 0.30 million, then falls to 1.452, which is why the gain is slightly below 1.528. The extra million becomes NATURE (4.2 to 4.5), PROMO (0.3 to 0.6) and the first INFRA in the region (0.4), taken mainly from Valparaíso. The same office in Aysén adds 1.46. The office pays for itself if it costs less than about USD 1.47 million over three years, a bar a lean office (USD 0.2 to 0.3 million a year) clears 1.6 to 2.5 times. **Recommendation: fund the Magallanes office first, from outside the Fund, then Aysén.**

5. Connecting the two decisions (Mandate C)

C-1. Yes: move USD 1 million of Magallanes' allocation into INFRA. *Cost:* forcing ALLOC (MAGALLANES, INFRA) ≥ 1.0 lowers Model A from 121.5194 to 121.3732, a loss of USD 0.146 million. The reduced cost (0.132) holds only for small amounts; near 1.0 the forcing row's dual is 0.204 (valid from 0.857 to 1.20), which is why the actual cost is higher. Magallanes NATURE falls from 4.2 to 3.3 and PROMO from 0.3 to 0.2, and Arica takes 1.2 of NATURE to keep the national NATURE minimum; a literal NATURE for INFRA swap inside Magallanes would break that minimum, so 0.146 is the cost to quote. *Benefit:* the Magallanes bed dual is USD 430 per slot, the highest in Chile, valid for 900 slots; a re-solve with 1,000 more slots gives +USD 429,500 a season, so 2028 to 2030 is worth USD 1.29 million (1.11 million discounted at 8%): **8.8 to 1**, with breakeven near 114 slots a year. Paired with the A-7 office, the move costs only USD 0.090 million (122.9943 to 122.9043). *Assumptions:* 1,000 slots per USD 1 million from 2028 (engineers' estimate); 2027 seats, spending and costs repeat in 2028 to 2030; Model A measures gross spending over three years and Model B net contribution per season, a units gap that the nine to one ratio comfortably absorbs; the NATURE money taken from park management is a real but unmodeled cost. **Decision:** move the money, contract it as beds rather than general infrastructure, fund the office, and re-solve before going beyond 900 slots, when Punta Arenas seats become the limit. Aysén (USD 345 per bed, valid to 7,700 slots) is the next candidate.

C-2. What the models do not see. *Why the money comes in blocks.* Each multiplier is a constant, so the hundredth dollar in a program earns exactly what the first did. A linear objective is maximized at a corner of the feasible region, where only as many variables are positive as there are binding rules, so each dollar flows to the best remaining use until a rule stops it. That is why 51 of 75 cells are zero, why Los Lagos and Araucanía are 100% TALENT and Aysén 100% INFRA, and why small values such as Coquimbo's 0.01 of SMART appear where a rule binds exactly. Real programs saturate (the tenth training course in a region fills less than the first), so constant returns overstate concentration and every dual price holds only over a short range (the Fund +0.31, the Magallanes ceiling +0.30). *What else is missing:*

- **(1) Uncertainty.** The multipliers are point estimates, and in RfP 2 our own forecasts lost to a naive seasonal rule in 26 of 45 tests. A 10% error in a few multipliers can change which program wins in a region, so the plan should be read as a ranking of choices, not a forecast.

- **(2) Carrying capacity.** Neither model limits park pressure, yet DB4 trail sensors already show non-open readings 2 times in 6 at Chiloé and 4 in 7 at Rapa Nui, and B-5 would add 5,500 visitors to Los Lagos.
- **(3) Community, indigenous and political perspectives.** Mapuche communities in Araucanía and Aymara communities in Arica y Parinacota carry no weight in either objective, or Metropolitana's 1.79 million (against our 6.5) will be politically hard to defend.
- **(4) Time and the link between the models.** INFRA creates no Model B beds before 2028, and the two models were joined by hand in C-1. Model B also covers only one season of international program visitors.

Phase 3 extension. One linked Fund and Season model for 2027 to 2030: Model A's INFRA in Aysén and Magallanes raises Model B's headroom in later seasons ($HCAP(r,t) = HCAP0(r) + k(r) \times INFRA(r)$, with $k \approx 1{,}000$ slots per USD 1 million), multipliers become piecewise linear with diminishing returns, Punta Arenas and Puerto Montt seats become purchasable decisions, parks get DB4 based caps, and the model is solved under low, base and high multiplier scenarios. C-1 type trade-offs then become decisions the model makes itself.

C-3 (bonus). Requiring each PROMO campaign to be zero or at least USD 1.0 million, with at most four campaigns, costs USD 0.48 million (121.0379). Antofagasta, Coquimbo, Valparaíso and Maule keep their campaigns; Atacama loses its own, because its promotion money loses least when it moves to INFRA (2.88 against 2.30), and Magallanes' USD 0.3 million seed also disappears. Dual prices from this integer version describe the plan with the campaign choices frozen and cannot capture the jump when a campaign switches on or off, so every sensitivity question there must be re-solved.

6. Risks, limits and next steps

Risk	What the models say	How we manage it
Multipliers are wrong	Most dual prices hold over narrow ranges; small coefficient changes can switch a program.	Read the plan as a ranking; re-solve every policy change; Phase 3 scenarios.
Patagonia cannot execute	Magallanes and Aysén ceilings carry the highest duals (1.53 and 1.47).	Project offices, Magallanes first; disburse early in the fiscal year.
Airport closure	Puerto Montt at half capacity costs USD 2.45 million a season.	Santiago to Balmaceda contingency; Puerto Montt to Punta Arenas bundle.
Headroom estimate	HCAP comes from one quarter (Q1 2024); on Q1 2025 data Magallanes headroom is 8,800, not 10,600.	Size the Punta Arenas seat request on Q1 2026 occupancy; ask for options, not firm seats.
Crowding and consent	Not modeled.	Park caps from DB4 sensors; consult communities before INFRA contracts.

What we ask the Vice Minister to approve, and when

1. The Fund (before DIPRES closes the budget): the USD 45 million model plan, with USD 1 million of Magallanes' allocation moved into accommodation INFRA and a lean Magallanes project office funded outside the Fund (Aysén next).

2. The summer (before airline seat contracts are signed): the Model B routing plan; negotiate about 2,600 extra Punta Arenas seats first, and agree a Santiago to Balmaceda contingency for Puerto Montt.

3. The rules: a USD 2.0 million equity floor earmarked to TALENT or INFRA, and the 80% decentralization share with Chiloé safeguards; hold the jobs target at 1,600 and the SMART minimum at 3.0 until their wider benefits are measured.

4. Phase 3 (2027): our firm collects cost per bed and park sensor data during the season and builds the linked Fund and Season model for the 2028 budget.

Limits of the models. Both models are linear, deterministic and solved separately: they assume constant returns, certain multipliers, one season of visitors and no link between money spent and beds created (C-2). The numbers are therefore most reliable as comparisons between options under today's rules, and least reliable as forecasts of what Chile will actually earn. That is why every recommendation above is paired with a re-solve trigger and a data step.

What our firm will do next

When	What our firm does	Decision it supports
October to November 2026	Re-solve the approved package (floor 2.0, Magallanes INFRA, project office) as one model and brief DIPRES; prepare the seat request with SERNATUR.	Budget closure; airline contracts
December 2026	Update headroom with Q1 2026 occupancy; confirm the Punta Arenas seat number and the Balmaceda contingency.	Final seat and bed commitments
January to March 2027	Track routed visitors, occupancy and DB4 trail sensors weekly against the plan; flag any region near 88% occupancy.	In-season rerouting
April to June 2027	Re-estimate multipliers with 2027 data; build and test the linked Phase 3 model.	The 2028 Fund

7. What the plan means for each region

Each governor will ask what happens to their region: one line each, combining the Fund plan (Model A) and the summer plan (Model B, "full" = beds full).

Region	Fund 2027 (USD M)	Visitors (k)	What it means for the region
Arica y Parinacota	1.20: TALENT 0.90, NATURE 0.30	4.9	Wait tier: floor money builds skills first.
Tarapacá	1.20: TALENT 0.34, INFRA 0.86	0.9 (min)	Foundations first; summer at the promised minimum.
Antofagasta	4.50: INFRA 2.40, PROMO 2.10	6.2 (full)	Full ceiling; promotion matched by infrastructure.
Atacama	3.00: INFRA 1.54, PROMO 1.46	15.5 (full)	Full ceiling; first campaign lost if campaigns must be USD 1 M.
Coquimbo	4.00: TALENT 2.11, PROMO 1.88	9.2 (full)	Full ceiling; our Scale Now call confirmed.
Valparaíso	4.13: TALENT 2.19, PROMO 1.94	20.5	Gives 5,500 visitors to Los Lagos at 80% decentralization.
Metropolitana	1.79: SMART 1.79	7.0 (full)	Smart platforms, not promotion; co-fund campaigns privately.
O'Higgins	1.20: INFRA 1.20	1.0 (min)	Floor region; rises to USD 2.0 M under the new floor.
Maule	3.50: TALENT 1.83, PROMO 1.67	6.3 (full)	Full ceiling; keeps its campaign under C-3.
Biobío	1.20: SMART 1.20	13.5 (full)	Floor region; rises to USD 2.0 M under the new floor.
Araucanía	4.00: TALENT 4.00	1.5 (min)	Full ceiling; the summer needs higher spending per visitor.
Los Ríos	2.78: TALENT 2.78	1.1 (min)	Gives USD 0.78 M to floor regions if the floor rises.
Los Lagos	5.00: TALENT 5.00	9.8	Gains 5,500 visitors at 80%; most exposed to a PMC closure.
Aysén	3.00: INFRA 3.00	2.0 (full)	Full ceiling; second project office; beds are the limit.
Magallanes	4.50: NATURE 4.20, PROMO 0.30	10.6 (full)	USD 1 M moves to beds; first project office; more PUQ seats.

Appendix A: Formulation of both models

Model A: 2027 Regional Tourism Development Fund

In words. Decide how many USD millions each of 15 regions receives in each of 5 programs (TALENT, INFRA, PROMO, SMART, NATURE) to maximize projected incremental visitor spending in 2027 to 2029, subject to the Fund, an equity floor, absorption ceilings, foundations first, digital readiness, national program rules, the jobs commitment and macrozone balance.

In math.

Row name in LINGO	Mathematical form (r in REGION, p in PROGRAM)	Rows
Decision variables	ALLOC(r,p) ≥ 0, USD million	75 vars
TOTAL_SPENDING (objective)	max Σr Σp MULT(r,p) × ALLOC(r,p)	–
TOTAL_BUDGET	Σr Σp ALLOC(r,p) ≤ 45.0	1
REGION_MINIMUM(r)	Σp ALLOC(r,p) ≥ 1.2	15
REGION_ABSORPTION(r)	Σp ALLOC(r,p) ≤ ABSORB(r)	15
REGION_PROMO_LIMIT(r)	ALLOC(r,PROMO) ≤ 0.75 × [ALLOC(r,TALENT) + ALLOC(r,INFRA)] + 0.3	15
REGION_SMART_LIMIT(r)	ALLOC(r,SMART) ≤ SMARTCAP(r) = 0.5 × AI index(r)	15
NATIONAL_TALENT / INFRA / SMART / NATURE_MIN, NATIONAL_PROMO_MAX	Σr ALLOC(r,p) ≥ 9.0, 9.0, 3.0, 4.5; Σr ALLOC(r,PROMO) ≤ 11.25	5
TOTAL_JOBS_MIN	Σr Σp JOBS(r,p) × ALLOC(r,p) ≥ 1,600	1
NORTE_MIN, SUR_MIN, AUSTRAL_MIN, CENTRO_MAX	Σ(r in NORTE) Σp ALLOC ≥ 9.0, NORTE = {ARICA, TARAPACA, ANTOFAGASTA, ATACAMA, COQUIMBO}; Σ(r in SUR) ≥ 11.25, SUR = {BIOBIO, ARAUCANIA, LOSRIOS, LOSLAGOS}; Σ(r in AUSTRAL) ≥ 4.5, AUSTRAL = {AYSEN, MAGALLANES}; Σ(r in CENTRO) ≤ 15.75, CENTRO = {VALPARAISO, METROPOLITANA, OHIGGINS, MAULE}	4
Total	**75 variables, 71 constraints (LINGO reports 72 rows, counting the objective)**	**71**

Model B: High Season 2027

In words. Decide how many thousand program visitors to route from each of 6 gateways to each of 15 regions to maximize net contribution (in-region spending minus connectivity cost), subject to gateway seats, spare beds at an 88% occupancy ceiling, committed minimums and 75% decentralization.

In math.

Row name in LINGO	Mathematical form (g in GATEWAY, r in REGION)	Rows
Decision variables	Y(g,r) ≥ 0, thousand visitors	90 vars
TOTAL_NET_CONTRIBUTION (objective)	max Σg Σr [SPEND(r) − COST(g,r)] × Y(g,r), USD thousand	–
GATEWAY_SEATS(g)	Σr Y(g,r) ≤ GCAP(g)	6
REGION_HEADROOM(r)	Σg Y(g,r) ≤ HCAP(r)	15
REGION_COMMITMENT(r)	Σg Y(g,r) ≥ MINV(r)	15
DECENTRALIZATION	Σg Σ(r in OUTSIDE_CORE) Y(g,r) ≥ 0.75 × Σg Σr Y(g,r); OUTSIDE_CORE = the 13 regions other than VALPARAISO and METROPOLITANA	1
Total	**90 variables, 37 constraints (LINGO reports 38 rows). COST entered with gateways as rows.**	**37**

C-3 extension (bonus)

CAMPAIGN(r) ∈ {0,1}; PROMO_MIN_SIZE(r): ALLOC(r,PROMO) ≥ 1.0 × CAMPAIGN(r); PROMO_SWITCH(r): ALLOC(r,PROMO) ≤ ABSORB(r) × CAMPAIGN(r); MAX_CAMPAIGNS: Σr CAMPAIGN(r) ≤ 4. 90 variables (15 integer), 103 rows.

Data lineage check (SPEND and HCAP against DB3, Q1 2024)

SPEND = stay × (ADR ÷ 1.5 + 90), rounded to USD 10. HCAP = (arrivals × 88 ÷ occupancy % − arrivals) ÷ 1,000. All 15 regions reproduce Exhibit B2 exactly (Supporting_Workbook.xlsx in the zip, tab Lineage_DB3, live formulas). Magallanes: 4.39 × (200.73 ÷ 1.5 + 90) = 982.6, so USD 980; arrivals 47,470 at 71.9% occupancy give 58,100 − 47,470 = 10.6 thousand. Aysén: 4.44 × (152.58 ÷ 1.5 + 90) = 851.2, so USD 850; arrivals 22,590 at 80.9% give 2.0 thousand. Both match.

Appendix B: LINGO code for both base models

Exact text of ModelA_Base.lng and ModelB_Base.lng in the zip. Every scenario file changes one line (marked CHANGED) or adds one row at the end.

ModelA_Base.lng

```
MODEL:
! TGM 530 Final Project Part 2 - Model A: 2027 Regional Tourism Development Fund;
! Prolific Consulting (Group 1). Base model: 75 variables, 71 constraints (LINGO shows 72 rows);
! Units: ALLOC in USD million, objective = projected incremental visitor spending 2027-2029 in USD million;
! Base model or scenario: the file name names the scenario. A changed line is marked CHANGED, an added row is at the end;
SETS:
 REGION / ARICA TARAPACA ANTOFAGASTA ATACAMA COQUIMBO VALPARAISO METROPOLITANA OHIGGINS MAULE
          BIOBIO ARAUCANIA LOSRIOS LOSLAGOS AYSEN MAGALLANES /: ABSORB, SMARTCAP;
 PROGRAM / TALENT INFRA PROMO SMART NATURE /;
 NORTE(REGION)    / ARICA TARAPACA ANTOFAGASTA ATACAMA COQUIMBO /;
 CENTRO(REGION)   / VALPARAISO METROPOLITANA OHIGGINS MAULE /;
 SUR(REGION)      / BIOBIO ARAUCANIA LOSRIOS LOSLAGOS /;
 AUSTRAL(REGION)  / AYSEN MAGALLANES /;
 PLAN(REGION, PROGRAM): MULT, JOBS, ALLOC;
ENDSETS

DATA:
! Absorption ceiling CAPR, USD M (Exhibit A1);
 ABSORB = 2.5 2.5 4.5 3 4 5 4 2.5 3.5 4 4 3 5 3 4.5;
! SMART cap = 0.5 x AI & analytics maturity index (Exhibit A1, DB5), USD M;
 SMARTCAP = 1.36 1.095 1.385 1.125 1.59 1.36 1.79 0.505 1.36 1.335 0.57 0.645 1.195 0.975 1.475;
! Spending multipliers 2027-2029 (Exhibit A2): TALENT INFRA PROMO SMART NATURE;
 MULT =
    2.24 2.03 2.34 1.49 2.18
    2.00 1.85 2.29 1.53 1.57
    2.51 2.48 3.17 1.93 2.46
    2.34 2.30 2.88 1.55 2.28
    2.45 2.11 3.07 2.10 2.03
    2.15 1.85 3.08 1.83 1.52
    2.07 1.55 2.65 2.15 1.23
    1.85 1.78 1.96 1.08 1.27
    2.39 2.20 3.01 1.81 1.72
    2.07 1.95 2.38 1.77 1.63
    2.37 2.04 2.28 1.08 2.09
    2.21 1.85 2.29 1.30 2.15
    2.53 2.07 2.92 1.71 2.46
    3.68 3.73 3.27 1.79 3.55
    3.74 3.69 4.35 2.90 3.88;
! Direct jobs per USD 1 million (Exhibit A3);
 JOBS =
    47 40 23 13 32
    45 38 22 12 30
    38 32 19 10 26
    45 38 22 12 30
    45 38 22 12 30
    43 36 21 11 28
    36 30 18 10 24
    54 46 26 14 36
    50 42 24 13 33
    45 38 22 12 30
    54 46 26 14 36
    52 44 25 14 34
    45 38 22 12 30
    40 34 20 11 27
    38 32 19 10 26;
ENDDATA

! Objective: maximize projected incremental visitor spending (USD M);
[TOTAL_SPENDING] MAX = @SUM(PLAN(R, P): MULT(R, P) * ALLOC(R, P));

! 1. The Fund;
[TOTAL_BUDGET] @SUM(PLAN(R, P): ALLOC(R, P)) <= 45.0;
! 2. Regional equity floor;
@FOR(REGION(R): [REGION_MINIMUM] @SUM(PROGRAM(P): ALLOC(R, P)) >= 1.2);
! 3. Absorption ceiling;
@FOR(REGION(R): [REGION_ABSORPTION] @SUM(PROGRAM(P): ALLOC(R, P)) <= ABSORB(R));
! 4. Foundations first: PROMO <= 0.75 x (TALENT + INFRA) + 0.3 seed;
@FOR(REGION(R): [REGION_PROMO_LIMIT]
     ALLOC(R, @INDEX(PROGRAM, PROMO)) <=
     0.75 * (ALLOC(R, @INDEX(PROGRAM, TALENT)) + ALLOC(R, @INDEX(PROGRAM, INFRA))) + 0.3);
! 5. Digital readiness;
@FOR(REGION(R): [REGION_SMART_LIMIT] ALLOC(R, @INDEX(PROGRAM, SMART)) <= SMARTCAP(R));

! 6. National program commitments;
[NATIONAL_TALENT_MIN] @SUM(REGION(R): ALLOC(R, @INDEX(PROGRAM, TALENT))) >= 9.0;
[NATIONAL_INFRA_MIN]  @SUM(REGION(R): ALLOC(R, @INDEX(PROGRAM, INFRA)))  >= 9.0;
[NATIONAL_SMART_MIN]  @SUM(REGION(R): ALLOC(R, @INDEX(PROGRAM, SMART)))  >= 3.0;
[NATIONAL_NATURE_MIN] @SUM(REGION(R): ALLOC(R, @INDEX(PROGRAM, NATURE))) >= 4.5;
[NATIONAL_PROMO_MAX]  @SUM(REGION(R): ALLOC(R, @INDEX(PROGRAM, PROMO)))  <= 11.25;

! 7. Jobs commitment;
[TOTAL_JOBS_MIN] @SUM(PLAN(R, P): JOBS(R, P) * ALLOC(R, P)) >= 1600;

! 8. Macrozone balance;
[NORTE_MIN]    @SUM(NORTE(R):    @SUM(PROGRAM(P): ALLOC(R, P))) >= 9.0;
[SUR_MIN]      @SUM(SUR(R):      @SUM(PROGRAM(P): ALLOC(R, P))) >= 11.25;
[AUSTRAL_MIN]  @SUM(AUSTRAL(R):  @SUM(PROGRAM(P): ALLOC(R, P))) >= 4.5;
[CENTRO_MAX]   @SUM(CENTRO(R):   @SUM(PROGRAM(P): ALLOC(R, P))) <= 15.75;

END
```

ModelB_Base.lng

```
MODEL:
! TGM 530 Final Project Part 2 - Model B: High Season 2027 (January-March);
! Prolific Consulting (Group 1). Base model: 90 variables, 37 constraints (LINGO shows 38 rows);
! Units: Y in thousand visitors, SPEND and COST in USD per visitor, objective in USD thousand;
! Base model or scenario: the file name names the scenario. A changed line is marked CHANGED, an added row is at the end;
SETS:
 GATEWAY / SCL NORTE PMC PUQ PASO CRUCERO /: GCAP;
 REGION / ARICA TARAPACA ANTOFAGASTA ATACAMA COQUIMBO VALPARAISO METROPOLITANA OHIGGINS MAULE
          BIOBIO ARAUCANIA LOSRIOS LOSLAGOS AYSEN MAGALLANES /: SPEND, HCAP, MINV;
 ! Regions that count toward decentralization (all except VALPARAISO and METROPOLITANA);
 OUTSIDE_CORE(REGION) / ARICA TARAPACA ANTOFAGASTA ATACAMA COQUIMBO OHIGGINS MAULE
                        BIOBIO ARAUCANIA LOSRIOS LOSLAGOS AYSEN MAGALLANES /;
 ROUTE(GATEWAY, REGION): COST, Y;
ENDSETS

DATA:
! Program seats, thousand visitors (Exhibit B1);
 GCAP = 55 12 12 8 14 9;
! In-region spending per visitor, USD (Exhibit B2, from DB3 Q1 2024);
 SPEND = 450 400 460 460 430 410 480 360 490 460 410 400 450 850 980;
! Spare accommodation capacity at 88% occupancy, thousand visitors (Exhibit B2);
 HCAP = 8.3 16.1 6.2 15.5 9.2 39.8 7 15 6.3 13.5 15.5 13.6 37.5 2 10.6;
! Committed minimum, thousand visitors (Exhibit B2);
 MINV = 1 0.9 1.6 1.1 1.4 2.7 2.8 1 1.2 1.4 1.5 1.1 2.1 0.7 1.7;
! Connectivity cost per visitor, USD. GATEWAYS AS ROWS (B3_ConnCost tab order);
 COST =
  135 115 100  60  40  40  25  30  50  45  70  70  70 160 170
   60  55  55  80  95 125 105 135 135 135 155 155 175 250 275
  205 190 165 145 130 105  95  80  85  65  65  40  45 100 150
  320 305 275 240 225 210 195 185 190 160 150 145 130 135  70
  215 175 135  80  55  15  25  40  60  65  95 110 135 240 300
  260 220 175 115  60  15  40  40  65  70 110 125 150 245 355;
ENDDATA

! Objective: maximize net economic contribution, USD thousand;
[TOTAL_NET_CONTRIBUTION] MAX = @SUM(ROUTE(G, R): (SPEND(R) - COST(G, R)) * Y(G, R));

! 1. Gateway seats;
@FOR(GATEWAY(G): [GATEWAY_SEATS] @SUM(REGION(R): Y(G, R)) <= GCAP(G));
! 2. Accommodation headroom;
@FOR(REGION(R): [REGION_HEADROOM] @SUM(GATEWAY(G): Y(G, R)) <= HCAP(R));
! 3. Regional commitments;
@FOR(REGION(R): [REGION_COMMITMENT] @SUM(GATEWAY(G): Y(G, R)) >= MINV(R));
! 4. Decentralization: share of visitors outside Metropolitana and Valparaiso;
[DECENTRALIZATION] @SUM(GATEWAY(G): @SUM(OUTSIDE_CORE(R): Y(G, R))) >= 0.75 * @SUM(ROUTE(G, R): Y(G, R));

END
```

Appendix C: LINGO Solution and Range reports, base models

LINGO 22.0 output. In the Solution reports, lines that only echo input data (ABSORB, MULT, SPEND, COST and so on) are omitted and runs of spaces are condensed to fit the page; values are unchanged. The full reports are in the zip.

Model A, Solution report (ModelA_Base_Solution.txt)

```
LINGO/WIN64 22.0.147 (24 Mar 2026), LINDO API 16.0.7099.134
Licensee info: Eval Use Only
License expires: 31 MAR 2027
Global optimal solution found.
Objective value:                              121.5194
Infeasibilities:                              0.000000
Total solver iterations:                             47
Time to best solution seconds:                     0.07
Elapsed runtime seconds:                           0.07
Model Class:                                         LP
Total variables:                75
Nonlinear variables:             0
Integer variables:               0
Total constraints:              77
Nonlinear constraints:           0
Total nonzeros:                585
Nonlinear nonzeros:              0
Variable           Value        Reduced Cost
ALLOC( ARICA, TALENT)                0.9000000                0.0000000
ALLOC( ARICA, INFRA)                 0.0000000      0.6000000E-01
ALLOC( ARICA, PROMO)                 0.0000000                0.3405302
ALLOC( ARICA, SMART)                 0.0000000                0.3337582
ALLOC( ARICA, NATURE)                0.3000000                0.0000000
ALLOC( TARAPACA, TALENT)             0.3428571                0.0000000
ALLOC( TARAPACA, INFRA)              0.8571429                0.0000000
ALLOC( TARAPACA, PROMO)              0.0000000                0.1131891
ALLOC( TARAPACA, SMART)              0.0000000      0.3538206E-01
ALLOC( TARAPACA, NATURE)             0.0000000                0.3700000
ALLOC( ANTOFAGASTA, TALENT)          0.0000000                0.1158837
ALLOC( ANTOFAGASTA, INFRA)           2.400000                 0.0000000
ALLOC( ANTOFAGASTA, PROMO)           2.100000                 0.0000000
ALLOC( ANTOFAGASTA, SMART)           1.500000                 0.1158837
ALLOC( ANTOFAGASTA, NATURE)          0.0000000                0.2472093
ALLOC( ATACAMA, TALENT)              0.0000000                0.1100000
ALLOC( ATACAMA, INFRA)               0.0000000                0.1158837
ALLOC( ATACAMA, PROMO)               1.357143                 0.0000000
ALLOC( ATACAMA, SMART)               0.0000000                0.4685714
ALLOC( ATACAMA, NATURE)              0.0000000                0.1131894
ALLOC( COQUIMBO, TALENT)             2.108571                 0.0000000
ALLOC( COQUIMBO, INFRA)              0.0000000                0.1900000
ALLOC( COQUIMBO, PROMO)              0.8814286                0.0000000
ALLOC( COQUIMBO, SMART)        0.1000000E-01                  0.0000000
ALLOC( COQUIMBO, NATURE)             0.0000000                0.4446179
ALLOC( VALPARAISO, TALENT)           0.0000000                0.1158837
ALLOC( VALPARAISO, INFRA)            0.0000000                0.1500000
ALLOC( VALPARAISO, PROMO)            1.342224                 0.0000000
ALLOC( VALPARAISO, SMART)            1.000000       0.9239780E-01
ALLOC( VALPARAISO, NATURE)           0.0000000                0.1953488
ALLOC( METROPOLITANA, TALENT)        0.0000000                0.4339815
ALLOC( METROPOLITANA, INFRA)         0.0000000                0.1158837
ALLOC( METROPOLITANA, PROMO)         0.0000000                0.1846512
ALLOC( METROPOLITANA, SMART)         1.790000                 0.0000000
ALLOC( METROPOLITANA, NATURE)        0.0000000                0.1158837
ALLOC( OHIGGINS, TALENT)             0.0000000      0.6162791E-01
ALLOC( OHIGGINS, INFRA)              1.200000                 0.0000000
ALLOC( OHIGGINS, PROMO)              0.0000000                0.4104465
ALLOC( OHIGGINS, SMART)              0.0000000                0.5256146
ALLOC( OHIGGINS, NATURE)             0.0000000                0.6367442
ALLOC( MAULE, TALENT)                1.828571                 0.0000000
ALLOC( MAULE, INFRA)                 0.0000000      0.5837209E-01
ALLOC( MAULE, PROMO)                 1.671429                 0.0000000
ALLOC( MAULE, SMART)                 0.0000000                0.7077409
ALLOC( MAULE, NATURE)                0.0000000                0.7077409
ALLOC( BIOBIO, TALENT)               0.0000000                0.1346179
ALLOC( BIOBIO, INFRA)                0.0000000                0.1346179
ALLOC( BIOBIO, PROMO)                0.0000000                0.2471761
ALLOC( BIOBIO, SMART)                1.200000                 0.0000000
ALLOC( BIOBIO, NATURE)               0.0000000                0.1346179
ALLOC( ARAUCANIA, TALENT)            4.000000                 0.0000000
ALLOC( ARAUCANIA, INFRA)             0.0000000                0.1941781
ALLOC( ARAUCANIA, PROMO)             0.0000000                0.1941781
ALLOC( ARAUCANIA, SMART)             0.0000000                0.9839867
ALLOC( ARAUCANIA, NATURE)            0.0000000                0.2711163
ALLOC( LOSRIOS, TALENT)              2.000000                 0.0000000
ALLOC( LOSRIOS, INFRA)               0.0000000                0.2283721
ALLOC( LOSRIOS, PROMO)               0.0000000                0.2471761
ALLOC( LOSRIOS, SMART)               0.0000000                0.7522255
ALLOC( LOSRIOS, NATURE)              0.0000000      0.5511628E-01
ALLOC( LOSLAGOS, TALENT)             0.0000000                0.3000000
ALLOC( LOSLAGOS, INFRA)              0.0000000                0.1900000
ALLOC( LOSLAGOS, PROMO)              0.0000000      0.3255814E-01
ALLOC( LOSLAGOS, SMART)              0.0000000      0.1000000E-01
ALLOC( LOSLAGOS, NATURE)             0.0000000      0.1000000E-01
ALLOC( AYSEN, TALENT)                0.0000000                0.2183721
ALLOC( AYSEN, INFRA)                 3.000000                 0.0000000
ALLOC( AYSEN, PROMO)                 0.0000000                0.9998140
ALLOC( AYSEN, SMART)                 0.0000000                1.600226
ALLOC( AYSEN, NATURE)                0.0000000                0.2742718
ALLOC( MAGALLANES, TALENT)           0.0000000                0.2507558
ALLOC( MAGALLANES, INFRA)            0.0000000                0.1323837
ALLOC( MAGALLANES, PROMO)      0.3000000E-01                  0.0000000
ALLOC( MAGALLANES, SMART)            0.0000000                0.5086379
ALLOC( MAGALLANES, NATURE)           4.200000                 0.0000000

Row    Slack or Surplus      Dual Price
TOTAL_SPENDING                121.5194                1.000000
TOTAL_BUDGET                  0.000000                3.165349
REGION_MINIMUM( ARICA)        0.000000      -0.6186047E-01
REGION_MINIMUM( TARAPACA)     3.300000                -0.386047
REGION_MINIMUM( ANTOFAGASTA)  3.300000                 0.000000
REGION_MINIMUM( ATACAMA)      1.800000                 0.000000
REGION_MINIMUM( COQUIMBO)     1.800000                 0.000000
REGION_MINIMUM( VALPARAISO)   2.933488                 0.000000
REGION_MINIMUM( METROPOLITANA)0.5800000                0.000000
REGION_MINIMUM( OHIGGINS)     0.000000                -0.7751479
REGION_MINIMUM( MAULE)        2.300000                 0.000000
REGION_MINIMUM( BIOBIO)       0.000000                -0.1339867
REGION_MINIMUM( ARAUCANIA)    2.800000                 0.000000
REGION_MINIMUM( LOSRIOS)      1.576512                 0.000000
REGION_MINIMUM( LOSLAGOS)     3.800000                 0.000000
REGION_MINIMUM( AYSEN)        1.800000                 0.000000
REGION_MINIMUM( MAGALLANES)   3.300000                 0.000000
REGION_ABSORPTION( ARICA)     1.300000                 0.000000
REGION_ABSORPTION( TARAPACA)  1.300000                 0.000000
REGION_ABSORPTION( ANTOFAGASTA)0.0000000               0.2531162
REGION_ABSORPTION( ATACAMA)   0.0000000               0.1145947
REGION_ABSORPTION( COQUIMBO)  0.0000000               0.1961133
REGION_ABSORPTION( VALPARAISO)0.8665116                0.000000
REGION_ABSORPTION( METROPOLITANA) 2.200000             0.000000
REGION_ABSORPTION( OHIGGINS)  1.200000                 0.000000
REGION_ABSORPTION( MAULE)     0.0000000                0.2042525
REGION_ABSORPTION( BIOBIO)    2.800000                 0.000000
REGION_ABSORPTION( ARAUCANIA) 0.0000000                0.1907240
REGION_ABSORPTION( LOSRIOS)   0.2234884                0.000000
REGION_ABSORPTION( LOSLAGOS)  0.0000000                0.1212953
REGION_ABSORPTION( AYSEN)     0.0000000                1.527907
REGION_ABSORPTION( MAGALLANES) 0.0000000               0.1212953
REGION_PROMO_LIMIT( ARICA)    0.1925000                0.000000
REGION_PROMO_LIMIT( TARAPACA) 0.0000000                0.000000
REGION_PROMO_LIMIT( ANTOFAGASTA) 0.0000000     0.9860465E-01
REGION_PROMO_LIMIT( ATACAMA)  0.0000000      0.4252492E-02
REGION_PROMO_LIMIT( COQUIMBO) 0.0000000                0.7299729
REGION_PROMO_LIMIT( VALPARAISO) 0.0000000              0.3004851
REGION_PROMO_LIMIT( METROPOLITANA) 0.1300000           0.000000
REGION_PROMO_LIMIT( OHIGGINS) 0.0000000                0.000000
REGION_PROMO_LIMIT( MAULE)    0.0000000      0.8132890E-01
REGION_PROMO_LIMIT( BIOBIO)   0.3300000                0.000000
REGION_PROMO_LIMIT( ARAUCANIA) 3.300000                0.000000
REGION_PROMO_LIMIT( LOSRIOS)  2.382384                 0.000000
REGION_PROMO_LIMIT( LOSLAGOS) 4.050000                 0.000000
REGION_PROMO_LIMIT( AYSEN)    0.0000000                0.000000
REGION_PROMO_LIMIT( MAGALLANES) 0.0000000      0.5813953E-02
REGION_SMART_LIMIT( ARICA)    1.085000                 0.000000
REGION_SMART_LIMIT( TARAPACA) 1.085000                 0.000000
REGION_SMART_LIMIT( ANTOFAGASTA) 1.385000              0.000000
REGION_SMART_LIMIT( ATACAMA)  1.145000                 0.000000
REGION_SMART_LIMIT( COQUIMBO) 1.585000                 0.000000
REGION_SMART_LIMIT( VALPARAISO) 1.360000               0.000000
REGION_SMART_LIMIT( METROPOLITANA) 0.4000000           0.000000
REGION_SMART_LIMIT( OHIGGINS) 0.5050000                0.000000
REGION_SMART_LIMIT( MAULE)    1.360000                 0.000000
REGION_SMART_LIMIT( BIOBIO)   0.1360000                0.000000
REGION_SMART_LIMIT( ARAUCANIA) 0.5770000               0.000000
REGION_SMART_LIMIT( LOSRIOS)  0.1645000                0.000000
REGION_SMART_LIMIT( LOSLAGOS) 0.0000000                0.000000
REGION_SMART_LIMIT( AYSEN)    0.1975000                0.000000
REGION_SMART_LIMIT( MAGALLANES) 1.475000               0.000000
NATIONAL_INFRA_MIN            0.000000                 0.000000
NATIONAL_INFRA_MIN            0.000000       -0.2786047
NATIONAL_SMART_MIN            0.000000       -1.040897
NATIONAL_NATURE_MIN           0.000000       -0.1953488
NATIONAL_PROMO_MAX            0.897076                 0.000000
TOTAL_JOBS_MIN-               0.000000      -0.1837209E-01
NORTE_MIN -                   4.726512                 0.000000
SUR_MIN                       4.726512                 0.000000
AUSTRAL_MIN                   3.000000                 0.000000
CENTRO_MAX                    5.126512                 0.000000
```

Model A, Range report (ModelA_Base_RangeReport.txt)

```
  Ranges in which the basis is unchanged:
Objective coefficient ranges:
Current        Allowable      Allowable
Variable       Coefficient    Increase      Decrease
ALLOC( ARICA, TALENT)          2.740000  0.5813953E-02  0.6000000E-01
ALLOC( ARICA, INFRA)           2.030000  0.6000000E-01        INFINITY
ALLOC( ARICA, PROMO)           2.340000       0.3337582        INFINITY
ALLOC( ARICA, SMART)           2.180000  0.7557478E-01  0.5813953E-02
ALLOC( ARICA, NATURE)          2.180000  0.7557478E-01  0.5813953E-02
ALLOC( TARAPACA, TALENT)       2.030000  0.6162791E-01  0.7442860E-02
ALLOC( TARAPACA, INFRA)        2.090000  0.6162791E-01  0.7442860E-02
ALLOC( TARAPACA, PROMO)        2.630000        INFINITY        INFINITY
ALLOC( TARAPACA, SMART)        1.530000  0.3538206E-01        INFINITY
ALLOC( TARAPACA, NATURE)       2.030000  0.3700000E-01        INFINITY
ALLOC( ANTOFAGASTA, TALENT)    2.740000  0.1158837E-01        INFINITY
ALLOC( ANTOFAGASTA, INFRA)     2.400000        INFINITY        INFINITY
ALLOC( ANTOFAGASTA, PROMO)     2.100000        INFINITY        INFINITY
ALLOC( ANTOFAGASTA, SMART)     1.500000  0.1158837E-01        INFINITY
ALLOC( ANTOFAGASTA, NATURE)    2.180000  0.2472093E-01        INFINITY
ALLOC( ATACAMA, TALENT)        2.740000  0.1100000E-01        INFINITY
ALLOC( ATACAMA, INFRA)         2.400000  0.1158837E-01        INFINITY
ALLOC( ATACAMA, PROMO)         1.357143        INFINITY        INFINITY
ALLOC( ATACAMA, SMART)         1.810000  0.4685714E-01        INFINITY
ALLOC( ATACAMA, NATURE)        2.070000  0.1131894E-01        INFINITY
ALLOC( COQUIMBO, TALENT)       2.108571        INFINITY        INFINITY
ALLOC( COQUIMBO, INFRA)        2.380000  0.1900000E-01        INFINITY
ALLOC( COQUIMBO, PROMO)        1.720000  0.1131894E-01        INFINITY
ALLOC( COQUIMBO, SMART)        1.710000        INFINITY  0.1961744E-01
ALLOC( COQUIMBO, NATURE)       2.370000  0.4446179E-01  0.1961744E-01
ALLOC( VALPARAISO, TALENT)     2.740000  0.1158837E-01        INFINITY
ALLOC( VALPARAISO, INFRA)      2.400000  0.1500000E-01        INFINITY
ALLOC( VALPARAISO, PROMO)      2.280000  0.5004186E-01        INFINITY
ALLOC( VALPARAISO, SMART)      1.500000  0.9239780E-01        INFINITY
ALLOC( VALPARAISO, NATURE)     2.090000  0.1953488E-01        INFINITY
ALLOC( METROPOLITANA, TALENT)  2.740000  0.4339815E-01        INFINITY
ALLOC( METROPOLITANA, INFRA)   2.400000  0.1158837E-01        INFINITY
ALLOC( METROPOLITANA, PROMO)   2.630000  0.1846512E-01        INFINITY
ALLOC( METROPOLITANA, SMART)   1.790000        INFINITY        INFINITY
ALLOC( METROPOLITANA, NATURE)  2.740000  0.1158837E-01        INFINITY
ALLOC( OHIGGINS, TALENT)       1.800000  0.6162791E-01  0.6162791E-01
ALLOC( OHIGGINS, INFRA)        1.200000        INFINITY        INFINITY
ALLOC( OHIGGINS, PROMO)        1.740000  0.4104465E-01        INFINITY
ALLOC( OHIGGINS, SMART)        1.200000  0.5256146E-01        INFINITY
ALLOC( OHIGGINS, NATURE)       1.200000  0.6367442E-01        INFINITY
ALLOC( MAULE, TALENT)          2.200000  0.5837209E-01  0.5837209E-01
ALLOC( MAULE, INFRA)           2.200000        INFINITY        INFINITY
ALLOC( MAULE, SMART)           1.810000  0.7557478E-01        INFINITY
ALLOC( MAULE, NATURE)          2.070000  0.1131894E-01        INFINITY
ALLOC( BIOBIO, TALENT)         2.070000  0.1346179E-01        INFINITY
ALLOC( BIOBIO, INFRA)          2.380000  0.1346179E-01        INFINITY
ALLOC( BIOBIO, PROMO)          1.720000  0.2471761E-01        INFINITY
ALLOC( BIOBIO, SMART)          1.710000        INFINITY  0.1961744E-01
ALLOC( BIOBIO, NATURE)         2.370000  0.1346179E-01  0.1961744E-01
ALLOC( ARAUCANIA, TALENT)      2.370000        INFINITY  0.1961744E-01
ALLOC( ARAUCANIA, INFRA)       2.030000  0.1941781E-01  0.1961744E-01
ALLOC( ARAUCANIA, PROMO)       2.280000  0.1941781E-01  0.1961744E-01
ALLOC( ARAUCANIA, SMART)       2.090000  0.9839867E-01        INFINITY
ALLOC( ARAUCANIA, NATURE)      2.090000  0.2711163E-01        INFINITY
ALLOC( LOSRIOS, TALENT)        1.830000  0.2608695E-01  0.4870129E-01
ALLOC( LOSRIOS, INFRA)         1.800000  0.2283721E-01        INFINITY
ALLOC( LOSRIOS, PROMO)         1.800000  0.2471761E-01        INFINITY
ALLOC( LOSRIOS, SMART)         1.530000  0.7522255E-01        INFINITY
ALLOC( LOSRIOS, NATURE)        2.130000  0.5511628E-01  0.1000000E-01
ALLOC( LOSLAGOS, TALENT)       2.070000  0.3000000E-01        INFINITY
ALLOC( LOSLAGOS, INFRA)        2.070000  0.1900000E-01        INFINITY
ALLOC( LOSLAGOS, PROMO)        2.325581  0.3255814E-01  0.3255814E-01
ALLOC( LOSLAGOS, SMART)        1.710000  0.1000000E-01        INFINITY
ALLOC( LOSLAGOS, NATURE)       1.710000  0.1000000E-01        INFINITY
ALLOC( AYSEN, TALENT)          3.280000  0.2183721E-01        INFINITY
ALLOC( AYSEN, INFRA)           3.000000  0.2742718E-01  0.2742718E-01
ALLOC( AYSEN, PROMO)           3.270000  0.9998140E-01        INFINITY
ALLOC( AYSEN, SMART)           3.200000  0.1600226E-01        INFINITY
ALLOC( AYSEN, NATURE)          3.200000  0.2742718E-01        INFINITY
ALLOC( MAGALLANES, TALENT)     3.740000  0.2507558E-01        INFINITY
ALLOC( MAGALLANES, INFRA)      3.690000  0.1323837E-01        INFINITY
ALLOC( MAGALLANES, PROMO)      4.350000  0.1762176E-01  0.5813953E-02
ALLOC( MAGALLANES, SMART)      3.880000  0.5086379E-01  0.1676418E-01
ALLOC( MAGALLANES, NATURE)     3.880000  0.5813953E-02  0.7564784E-01

Righthand Side Ranges:
Current        Allowable      Allowable
Row            RHS            Increase      Decrease
TOTAL_BUDGET                  25.00000       0.3070879      0.1226809
REGION_MINIMUM( ARICA)        1.200000        1.300000      0.3486521
REGION_MINIMUM( ANTOFAGASTA)  1.200000        3.300000        INFINITY
REGION_MINIMUM( ATACAMA)      1.200000        1.800000        INFINITY
REGION_MINIMUM( COQUIMBO)     1.200000        2.800000        INFINITY
REGION_MINIMUM( VALPARAISO)   1.200000        2.933488        INFINITY
REGION_MINIMUM( METROPOLITAN  1.200000        0.5800000        INFINITY
REGION_MINIMUM( OHIGGINS)     1.200000        2.300000      0.1339867
REGION_MINIMUM( MAULE)        1.200000        2.300000        INFINITY
REGION_MINIMUM( BIOBIO)       1.200000      0.1000000E-01     0.1339867
REGION_MINIMUM( ARAUCANIA)    1.200000        2.800000        INFINITY
REGION_MINIMUM( LOSRIOS)      1.200000        1.800000        INFINITY
REGION_MINIMUM( LOSLAGOS)     1.200000        3.800000        INFINITY
REGION_MINIMUM( AYSEN)        1.200000        1.800000        INFINITY
REGION_MINIMUM( MAGALLANES)   1.200000        3.300000        INFINITY
REGION_ABSORPTION( ARICA)     2.500000        INFINITY        1.300000
REGION_ABSORPTION( TARAPACA)  2.500000        INFINITY        1.300000
REGION_ABSORPTION( ANTOFAGASTA 4.500000       1.311455        0.6000000
REGION_ABSORPTION( ATACAMA)   3.000000        1.300000        0.6000000
REGION_ABSORPTION( COQUIMBO)  4.000000        3.206949        0.9422881
REGION_ABSORPTION( VALPARAIS  4.000000        INFINITY        0.8665116
REGION_ABSORPTION( METROPOLI  4.000000        INFINITY        2.200000
REGION_ABSORPTION( OHIGGINS)  2.500000        INFINITY        1.300000
REGION_ABSORPTION( MAULE)     4.000000        INFINITY        0.7751479
REGION_ABSORPTION( BIOBIO)    4.000000        INFINITY        2.800000
REGION_ABSORPTION( ARAUCANIA  4.000000        3.810458        0.2234884
REGION_ABSORPTION( LOSRIOS)   3.000000        INFINITY        0.2234884
REGION_ABSORPTION( LOSLAGOS)  3.000000        2.382384        0.2234884
REGION_ABSORPTION( AYSEN)     3.000000        0.8571429       0.8571429
REGION_ABSORPTION( MAGALLANE  4.000000        4.050000        0.3428571
REGION_PROMO_LIMIT( ARICA)    0.300000        INFINITY        0.9750000
REGION_PROMO_LIMIT( TARAPACA  0.300000        INFINITY        1.200000
REGION_PROMO_LIMIT( ANTOFAGA  0.300000        0.4861111       0.1300000
REGION_PROMO_LIMIT( ATACAMA)  0.300000        0.3133696       1.215000
REGION_PROMO_LIMIT( COQUIMBO  0.300000        0.3111940       1.215000
REGION_PROMO_LIMIT( VALPARAI  0.300000        0.1627126       0.1300000
REGION_PROMO_LIMIT( OHIGGINS  0.300000        INFINITY        1.200000
REGION_PROMO_LIMIT( METROPOL  0.300000        0.1771775       1.200000
REGION_PROMO_LIMIT( MAULE)    0.300000        INFINITY        0.1300000
REGION_PROMO_LIMIT( BIOBIO)   0.300000        INFINITY        1.200000
REGION_PROMO_LIMIT( ARAUCANI  0.300000        INFINITY        2.382384
REGION_PROMO_LIMIT( LOSRIOS)  0.300000        INFINITY        2.050000
REGION_PROMO_LIMIT( LOSLAGOS  0.300000        INFINITY        2.050000
REGION_PROMO_LIMIT( MAGALLAN  0.300000        0.1872076       0.1300000
REGION_SMART_LIMIT( ARICA)    1.085000        INFINITY        1.085000
REGION_SMART_LIMIT( TARAPACA  1.085000        INFINITY        1.085000
REGION_SMART_LIMIT( ANTOFAGA  1.385000        INFINITY        1.385000
REGION_SMART_LIMIT( ATACAMA)  1.145000        INFINITY        1.145000
REGION_SMART_LIMIT( COQUIMBO  1.585000        INFINITY        1.585000
REGION_SMART_LIMIT( VALPARAI  1.360000        INFINITY        1.360000
REGION_SMART_LIMIT( METROPOL  0.4000000       0.1771775       0.4000000
REGION_SMART_LIMIT( OHIGGINS  0.5050000     0.1000000E-01     0.5050000
REGION_SMART_LIMIT( MAULE)    1.360000        INFINITY        1.360000
REGION_SMART_LIMIT( BIOBIO)   0.1360000       INFINITY        0.1360000
REGION_SMART_LIMIT( ARAUCANI  0.5770000       INFINITY        0.5770000
REGION_SMART_LIMIT( LOSRIOS)  0.1645000       INFINITY        0.1645000
REGION_SMART_LIMIT( LOSLAGOS  0.1975000       INFINITY        0.1975000
REGION_SMART_LIMIT( MAGALLAN  1.475000        INFINITY        1.475000
NATIONAL_INFRA_MIN            9.000000        0.1428708       0.1000000E-01
NATIONAL_SMART_MIN            3.000000        0.1771775       0.1000000E-01
NATIONAL_NATURE_MIN           4.500000        INFINITY        1.897076
NATIONAL_PROMO_MAX            1.125000        INFINITY        0.1897076
TOTAL_JOBS_MIN-               9.000000        4.190697        4.190697
NORTE_MIN -                   9.000000        4.900000        INFINITY
SUR_MIN                       9.000000        4.900000        INFINITY
AUSTRAL_MIN                   4.500000        3.000000        INFINITY
CENTRO_MAX                    15.75000        INFINITY        5.126512
```

Model B, Solution report (ModelB_Base_Solution.txt)

```
LINGO/WIN64 22.0.147 (24 Mar 2026), LINDO API 16.0.7099.134
Licensee info: abande25@asu.edu
License expires: 28 MAR 2027
Global optimal solution found.
Objective value:                              50246.00
Infeasibilities:                              0.000000
Total solver iterations:                            35
Time to best solution seconds:                    0.15
Elapsed runtime seconds:                          0.15
Model Class:                                        LP
Total variables:                90
Nonlinear variables:             0
Integer variables:               0
Total constraints:              38
Nonlinear constraints:           0
Total nonzeros:                450
Nonlinear nonzeros:              0
Variable           Value        Reduced Cost
X( SCP, ARICA)                       0.000000                65.00000
X( SCP, TARAPACA)                    0.000000                50.00000
X( SCP, ANTOFAGASTA)                 0.000000                35.00000
X( SCP, ATACAMA)                     15.25000                0.000000
X( SCP, COQUIMBO)                    4.000000                0.000000
X( SCP, VALPARAISO)                  0.000000                25.00000
X( SCP, METROPOLITANA)               4.000000                0.000000
X( SCP, OHIGGINS)                    1.000000                0.000000
X( SCP, MAULE)                       6.300000                0.000000
X( SCP, BIOBIO)                      13.00000                0.000000
X( SCP, ARAUCANIA)                   13.50000                0.000000
X( SCP, LOSRIOS)                     0.000000                5.000000
X( SCP, LOSLAGOS)                    0.900000                0.000000
X( SCP, AYSEN)                       0.000000                35.00000
X( SCP, MAGALLANES)                  2.600000                0.000000
X( NORTE, ARICA)                     4.000000                0.000000
X( NORTE, TARAPACA)                  0.000000                0.000000
X( NORTE, ANTOFAGASTA)               6.200000                0.000000
X( NORTE, ATACAMA)                   0.000000                90.00000
X( NORTE, COQUIMBO)                  0.000000                65.00000
X( NORTE, VALPARAISO)                0.000000                120.00000
X( NORTE, METROPOLITANA)             0.000000                190.0000
X( NORTE, OHIGGINS)                  0.000000                115.0000
X( NORTE, MAULE)                     0.000000                95.00000
X( NORTE, BIOBIO)                    0.000000                100.0000
X( NORTE, ARAUCANIA)                 0.000000                95.00000
X( NORTE, LOSRIOS)                   0.000000                100.0000
X( NORTE, LOSLAGOS)                  0.000000                110.0000
X( NORTE, AYSEN)                     0.000000                135.0000
X( NORTE, MAGALLANES)                0.000000                115.0000
X( PMC, ARICA)                       0.000000                190.0000
X( PMC, TARAPACA)                    0.000000                160.0000
X( PMC, ANTOFAGASTA)                 0.000000                125.0000
X( PMC, ATACAMA)                     0.000000                110.0000
X( PMC, COQUIMBO)                    0.000000                115.0000
X( PMC, VALPARAISO)                  0.000000                115.0000
X( PMC, METROPOLITANA)               0.000000                280.0000
X( PMC, OHIGGINS)                    0.000000                75.00000
X( PMC, MAULE)                       0.000000                60.00000
X( PMC, BIOBIO)                      0.000000                50.00000
X( PMC, ARAUCANIA)                   0.000000                20.00000
X( PMC, LOSRIOS)                     1.100000                0.000000
X( PMC, LOSLAGOS)                    8.900000                0.000000
X( PMC, AYSEN)                       2.000000                0.000000
X( PMC, MAGALLANES)                  0.000000                0.000000
X( PUC, ARICA)                       0.000000                350.0000
X( PUC, TARAPACA)                    0.000000                340.0000
X( PUC, ANTOFAGASTA)                 0.000000                310.0000
X( PUC, ATACAMA)                     0.000000                280.0000
X( PUC, COQUIMBO)                    0.000000                285.0000
```

```
Y( PUQ, VALPARAISO)                     0.000000        295.0000
Y( PUQ, METROPOLITANA)                  0.000000        270.0000
Y( PUQ, OHIGGINS)                       0.000000        250.0000
Y( PUQ, MAULE)                          0.000000        240.0000
Y( PUQ, BIOBIO)                         0.000000        175.0000
Y( PUQ, ARAUCANIA)                      0.000000        180.0000
Y( PUQ, LOSRIOS)                        0.000000        180.0000
Y( PUQ, LOSLAGOS)                       0.000000        160.0000
Y( PUQ, AYSEN)                          0.000000        110.0000
Y( PUQ, MAGALLANES)                     8.000000        0.000000
Y( PASO, ARICA)                         0.000000        145.0000
Y( PASO, TARAPACA)                      0.000000        110.0000
Y( PASO, ANTOFAGASTA)                   0.000000        70.00000
Y( PASO, ATACAMA)                       0.000000        45.00000
Y( PASO, COQUIMBO)                      0.000000        15.00000
Y( PASO, VALPARAISO)                    11.50000        0.000000
Y( PASO, METROPOLITANA)                 2.500000        0.000000
Y( PASO, OHIGGINS)                      0.000000        10.00000
Y( PASO, MAULE)                         0.000000        10.00000
Y( PASO, BIOBIO)                        0.000000        25.00000
Y( PASO, ARAUCANIA)                     0.000000        25.00000
Y( PASO, LOSRIOS)                       0.000000        45.00000
Y( PASO, LOSLAGOS)                      0.000000        115.0000
Y( PASO, AYSEN)                         0.000000        115.0000
Y( PASO, MAGALLANES)                    0.000000        130.0000
Y( CRUCERO, ARICA)                      0.000000        190.0000
Y( CRUCERO, TARAPACA)                   0.000000        155.0000
Y( CRUCERO, ANTOFAGASTA)                0.000000        110.0000
Y( CRUCERO, ATACAMA)                    0.000000        55.00000
Y( CRUCERO, COQUIMBO)                   0.000000        20.00000
Y( CRUCERO, VALPARAISO)                 9.000000        0.000000
Y( CRUCERO, METROPOLITANA)              0.000000        15.00000
Y( CRUCERO, OHIGGINS)                   0.000000        10.00000
Y( CRUCERO, MAULE)                      0.000000        15.00000
Y( CRUCERO, BIOBIO)                     0.000000        25.00000
Y( CRUCERO, ARAUCANIA)                  0.000000        40.00000
Y( CRUCERO, LOSRIOS)                    0.000000        60.00000
Y( CRUCERO, LOSLAGOS)                   0.000000        80.00000
Y( CRUCERO, AYSEN)                      0.000000        120.0000
Y( CRUCERO, MAGALLANES)                 0.000000        185.0000

Row    Slack or Surplus      Dual Price
TOTAL NET CONTRIBUTION          50246.00        1.000000
GATEWAY_SEATS( SCL)             0.000000        383.7500
GATEWAY_SEATS( NORTE)           0.000000        398.7500
GATEWAY_SEATS( PMC)             0.000000        408.7500
GATEWAY_SEATS( PUQ)             0.000000        483.7500
GATEWAY_SEATS( PASO)            0.000000        383.7500
GATEWAY_SEATS( CRUCERO)         0.000000        383.7500
REGION_HEADROOM( ARICA)         3.400000        0.000000
REGION_HEADROOM( TARAPACA)      15.20000        0.000000
REGION_HEADROOM( ANTOFAGASTA)   0.000000        15.00000
REGION_HEADROOM( ATACAMA)       0.000000        20.00000
REGION_HEADROOM( COQUIMBO)      0.000000        10.00000
REGION_HEADROOM( VALPARAISO)    19.00000        0.000000
REGION_HEADROOM( METROPOLITANA) 0.100000        60.00000
REGION_HEADROOM( OHIGGINS)      14.00000        0.000000
REGION_HEADROOM( MAULE)         0.000000        10.00000
REGION_HEADROOM( BIOBIO)        0.100000        35.00000
REGION_HEADROOM( ARAUCANIA)     12.50000        0.000000
REGION_HEADROOM( LOSRIOS)       12.50000        0.000000
REGION_HEADROOM( LOSLAGOS)      27.70000        0.000000
REGION_HEADROOM( AYSEN)         0.000000        345.0000
REGION_HEADROOM( MAGALLANES)    0.000000        430.0000
REGION_COMMITMENT( ARICA)       3.800000        0.000000
REGION_COMMITMENT( TARAPACA)    4.000000        -45.00000
REGION_COMMITMENT( ANTOFAGASTA) 4.000000        0.000000
REGION_COMMITMENT( ATACAMA)     14.40000        0.000000
REGION_COMMITMENT( COQUIMBO)    7.800000        0.000000
REGION_COMMITMENT( VALPARAISO)  17.80000        0.000000
REGION_COMMITMENT( METROPOLITANA) 4.200000      0.000000
REGION_COMMITMENT( OHIGGINS)    0.000000        -50.00000
REGION_COMMITMENT( MAULE)       12.10000        0.000000
REGION_COMMITMENT( BIOBIO)      12.10000        0.000000
REGION_COMMITMENT( ARAUCANIA)   0.000000        -40.00000
REGION_COMMITMENT( LOSRIOS)     0.000000        -45.00000
REGION_COMMITMENT( LOSLAGOS)    7.700000        0.000000
REGION_COMMITMENT( AYSEN)       1.700000        0.000000
REGION_COMMITMENT( MAGALLANES)  8.900000        0.000000
DECENTRALIZATION                0.000000        -15.00000
```

Model B, Range report (ModelB_Base_RangeReport.txt)

```
 Ranges in which the basis is unchanged:
Objective coefficient ranges:
Current       Allowable       Allowable
Variable       Coefficient      Increase          Decrease
Y( SCL, ARICA)                  215.0000        60.00000        INFINITY
Y( SCL, TARAPACA)               285.0000        30.00000        INFINITY
Y( SCL, ANTOFAGASTA)            350.0000        INFINITY        20.00000
Y( SCL, ATACAMA)                400.0000        INFINITY        20.00000
Y( SCL, COQUIMBO)               390.0000        INFINITY        10.00000
Y( SCL, VALPARAISO)             370.0000        25.00000        INFINITY
Y( SCL, METROPOLITANA)          445.0000        10.00000        10.00000
Y( SCL, OHIGGINS)               330.0000        50.00000        10.00000
Y( SCL, MAULE)                  415.0000        INFINITY        20.00000
Y( SCL, BIOBIO)                 415.0000        INFINITY        20.00000
Y( SCL, ARAUCANIA)              340.0000        40.00000        20.00000
Y( SCL, LOSRIOS)                380.0000        5.000000        INFINITY
Y( SCL, LOSLAGOS)               380.0000        5.000000        5.000000
Y( SCL, AYSEN)                  690.0000        35.00000        INFINITY
Y( SCL, MAGALLANES)             810.0000        110.0000        5.000000
Y( NORTE, ARICA)                305.0000        30.00000        30.00000
Y( NORTE, TARAPACA)             345.0000        45.00000        50.00000
Y( NORTE, ANTOFAGASTA)          415.0000        INFINITY        70.00000
Y( NORTE, ATACAMA)              380.0000        35.00000        INFINITY
Y( NORTE, COQUIMBO)             385.0000        65.00000        INFINITY
Y( NORTE, VALPARAISO)           385.0000        150.0000        INFINITY
Y( NORTE, METROPOLITANA)        375.0000        90.00000        INFINITY
Y( NORTE, OHIGGINS)             225.0000        115.0000        INFINITY
Y( NORTE, MAULE)                355.0000        95.00000        INFINITY
Y( NORTE, BIOBIO)               325.0000        100.0000        INFINITY
Y( NORTE, ARAUCANIA)            255.0000        95.00000        INFINITY
Y( NORTE, LOSRIOS)              275.0000        110.0000        INFINITY
Y( NORTE, LOSLAGOS)             275.0000        115.0000        INFINITY
Y( NORTE, AYSEN)                600.0000        115.0000        INFINITY
Y( NORTE, MAGALLANES)           720.0000        115.0000        INFINITY
Y( PMC, ARICA)                  275.0000        150.0000        INFINITY
Y( PMC, TARAPACA)               210.0000        190.0000        INFINITY
Y( PMC, ANTOFAGASTA)            295.0000        125.0000        INFINITY
Y( PMC, ATACAMA)                315.0000        110.0000        INFINITY
Y( PMC, COQUIMBO)               300.0000        115.0000        INFINITY
Y( PMC, VALPARAISO)             300.0000        115.0000        INFINITY
Y( PMC, METROPOLITANA)          385.0000        95.00000        INFINITY
Y( PMC, OHIGGINS)               280.0000        75.00000        INFINITY
Y( PMC, MAULE)                  405.0000        60.00000        INFINITY
Y( PMC, BIOBIO)                 405.0000        90.00000        INFINITY
Y( PMC, ARAUCANIA)              345.0000        20.00000        INFINITY
Y( PMC, LOSRIOS)                360.0000        45.00000        5.000000
Y( PMC, LOSLAGOS)               405.0000        5.000000        INFINITY
Y( PMC, AYSEN)                  750.0000        INFINITY        30.00000
Y( PMC, MAGALLANES)             830.0000        5.000000        INFINITY
Y( PUQ, ARICA)                  130.0000        350.0000        INFINITY
Y( PUQ, TARAPACA)               95.00000        330.0000        INFINITY
Y( PUQ, ANTOFAGASTA)            185.0000        310.0000        INFINITY
Y( PUQ, ATACAMA)                255.0000        280.0000        INFINITY
Y( PUQ, COQUIMBO)               250.0000        285.0000        INFINITY
Y( PUQ, VALPARAISO)             200.0000        295.0000        INFINITY
Y( PUQ, METROPOLITANA)          285.0000        270.0000        INFINITY
Y( PUQ, OHIGGINS)               175.0000        250.0000        INFINITY
Y( PUQ, MAULE)                  300.0000        240.0000        INFINITY
Y( PUQ, BIOBIO)                 300.0000        175.0000        INFINITY
Y( PUQ, ARAUCANIA)              200.0000        180.0000        INFINITY
Y( PUQ, LOSRIOS)                255.0000        180.0000        INFINITY
Y( PUQ, LOSLAGOS)               270.0000        160.0000        INFINITY
Y( PUQ, AYSEN)                  710.0000        110.0000        INFINITY
Y( PUQ, MAGALLANES)             910.0000        INFINITY        8.000000
Y( PASO, ARICA)                 230.0000        145.0000        INFINITY
Y( PASO, TARAPACA)              225.0000        110.0000        INFINITY
Y( PASO, ANTOFAGASTA)           380.0000        70.00000        INFINITY
Y( PASO, ATACAMA)               380.0000        20.00000        INFINITY
Y( PASO, COQUIMBO)              375.0000        15.00000        INFINITY
Y( PASO, VALPARAISO)            395.0000        11.50000        11.50000
Y( PASO, METROPOLITANA)         455.0000        2.500000        INFINITY
Y( PASO, OHIGGINS)              320.0000        10.00000        INFINITY
Y( PASO, MAULE)                 405.0000        10.00000        INFINITY
Y( PASO, BIOBIO)                395.0000        25.00000        INFINITY
Y( PASO, ARAUCANIA)             300.0000        25.00000        INFINITY
Y( PASO, LOSRIOS)               290.0000        45.00000        INFINITY
Y( PASO, LOSLAGOS)              315.0000        65.00000        INFINITY
Y( PASO, AYSEN)                 610.0000        115.0000        INFINITY
Y( PASO, MAGALLANES)            680.0000        130.0000        INFINITY
Y( CRUCERO, ARICA)              190.0000        190.0000        INFINITY
Y( CRUCERO, TARAPACA)           180.0000        175.0000        INFINITY
Y( CRUCERO, ANTOFAGASTA)        285.0000        110.0000        INFINITY
Y( CRUCERO, ATACAMA)            345.0000        55.00000        INFINITY
Y( CRUCERO, COQUIMBO)           335.0000        20.00000        INFINITY
Y( CRUCERO, VALPARAISO)         395.0000        INFINITY        18.00000
Y( CRUCERO, METROPOLITANA)      440.0000        15.00000        INFINITY
Y( CRUCERO, OHIGGINS)           325.0000        10.00000        INFINITY
Y( CRUCERO, MAULE)              405.0000        15.00000        INFINITY
Y( CRUCERO, BIOBIO)             390.0000        25.00000        INFINITY
Y( CRUCERO, ARAUCANIA)          300.0000        40.00000        INFINITY
Y( CRUCERO, LOSRIOS)            340.0000        60.00000        INFINITY
Y( CRUCERO, LOSLAGOS)           360.0000        80.00000        INFINITY
Y( CRUCERO, AYSEN)              705.0000        120.0000        INFINITY
Y( CRUCERO, MAGALLANES)         625.0000        185.0000        INFINITY
Righthand Side Ranges:
Current       Allowable       Allowable
Row             RHS           Increase          Decrease
GATEWAY_SEATS( SCL)             15.00000        10.00000        1.200000
GATEWAY_SEATS( NORTE)           12.00000        3.400000        8.900000
GATEWAY_SEATS( PMC)             8.000000        2.600000        8.900000
GATEWAY_SEATS( PUQ)             8.000000        2.600000        1.200000
GATEWAY_SEATS( PASO)            14.00000        6.000000        1.200000
GATEWAY_SEATS( CRUCERO)         9.000000        6.000000        1.200000
REGION_HEADROOM( ARICA)         8.400000        INFINITY        3.400000
REGION_HEADROOM( TARAPACA)      16.20000        INFINITY        15.20000
REGION_HEADROOM( ANTOFAGASTA    16.20000        3.000000        3.400000
REGION_HEADROOM( ATACAMA)       17.20000        0.900000        14.40000
REGION_HEADROOM( COQUIMBO)      9.200000        0.800000        7.800000
REGION_HEADROOM( VALPARAISO)    39.00000        INFINITY        19.00000
REGION_HEADROOM( METROPOLITA    7.000000        12.50000        2.500000
REGION_HEADROOM( OHIGGINS)      15.00000        INFINITY        14.00000
REGION_HEADROOM( MAULE)         6.300000        0.900000        5.100000
REGION_HEADROOM( BIOBIO)        13.10000        0.900000        12.10000
REGION_HEADROOM( ARAUCANIA)     13.50000        INFINITY        14.00000
REGION_HEADROOM( LOSRIOS)       37.80000        INFINITY        27.70000
REGION_HEADROOM( LOSLAGOS)      7.000000        INFINITY        2.300000
REGION_HEADROOM( MAGALLANES)    10.00000        0.900000        1.700000
REGION_COMMITMENT( ARICA)       10.00000        3.900000        INFINITY
REGION_COMMITMENT( TARAPACA)    0.900000        3.900000        0.900000
REGION_COMMITMENT( ANTOFAGAS    1.600000        4.160000        INFINITY
REGION_COMMITMENT( ATACAMA)     1.400000        14.40000        INFINITY
REGION_COMMITMENT( COQUIMBO)    1.400000        7.800000        INFINITY
REGION_COMMITMENT( VALPARAIS    2.700000        19.00000        INFINITY
REGION_COMMITMENT( METROPOLI    2.800000        4.200000        INFINITY
REGION_COMMITMENT( OHIGGINS)    1.200000        0.900000        INFINITY
REGION_COMMITMENT( MAULE)       1.200000        5.100000        INFINITY
REGION_COMMITMENT( BIOBIO)      1.200000        12.10000        INFINITY
REGION_COMMITMENT( ARAUCANIA    1.500000        0.900000        1.500000
REGION_COMMITMENT( LOSRIOS)     2.100000        7.700000        1.100000
REGION_COMMITMENT( LOSLAGOS)    2.200000        7.700000        INFINITY
REGION_COMMITMENT( AYSEN)       0.700000        1.300000        INFINITY
REGION_COMMITMENT( MAGALLANE    1.100000        8.900000        INFINITY
DECENTRALIZATION                0.000000        4.500000        0.9000000
```

Appendix D: Scenario log

#	Question	What changed · file	Predicted (dual or reduced cost × change)	Re-solved objective	Actual change	Exact?	Within range?	Decision insight
1	A-2	TOTAL_BUDGET 45 to 50 · ModelA_A2_Fund50	+15.8267	133.3571	+11.8377	No	No (to 45.307)	Diminishing returns: PROMO cap and four more ceilings bind.
2	A-2	TOTAL_BUDGET 45 to 40 · ModelA_A2_Fund40	−15.8267	104.1181	−17.4013	No	No (to 44.877)	Cuts cost more than the dual implies. Defend the 45.
3	A-3	TOTAL_JOBS_MIN 1,600 to 1,800 · ModelA_A3_Jobs1800	−3.6744	116.4086	−5.1108	No	No (to 1,604)	USD 25,600 per job on average. Advise against.
4	A-4	REGION_MINIMUM 1.2 to 2.0 (all 15) · ModelA_A4_Floor2	−0.6369	120.6534	−0.8660	No	n/a (15 rows)	Seven floors bind; extra Wait money goes to TALENT. Support 2.0.
5	A-5	NATIONAL_SMART_MIN 3.0 to 5.0 · ModelA_A5_Smart5	−2.0818	119.3474	−2.1720	No	No (to 3.178)	Costliest national rule; support only if spillovers exceed 2.2M.
6	A-6	Add ALLOC(METROPOLITANA, PROMO) ≥ 1.0 · ModelA_A6_MetroPromo1	−0.1847	120.9271	−0.5923	No	No (small amounts only)	Foundations first also forces 0.93 of TALENT; a full million costs 0.6M.
7	A-7	ABSORB(MAGALLANES) 4.5 to 5.5 · ModelA_A7_Capacity	+1.5279	122.9943	+1.4749	No	No (to +0.30)	Office pays for itself below about 1.47M over three years.
8	A-7	ABSORB(AYSEN) 3.0 to 4.0 · ModelA_A7_AysenCheck	+1.4679	122.9785	+1.4591	No	No (to +0.857)	Aysén is the second office.
9	B-2	GCAP(PUQ) 8 to 10 · ModelB_B2_Seats2	+967.5	51,213.5	+967.5	Yes	Yes (to +2.6)	Exact. Negotiate Punta Arenas first.
10	B-2	GCAP(PUQ) 8 to 12 · ModelB_B2_Seats4	+1,935.0	52,027.0	+1,781.0	No	No (to +2.6)	Beyond 2,600 seats the seat value falls to 373.75.
11	B-4	GCAP(PMC) 12 to 6 · ModelB_B4_PMC6	−2,452.5	47,793.5	−2,452.5	Yes	Yes (to −8.9)	Prepare a Balmaceda contingency.
12	B-5	Decentralization 0.75 to 0.80 · ModelB_B5_Dec80	≈ −82.5	50,153.5	−92.5	No	n/a (coefficient)	Valparaíso −5,500, Los Lagos +5,500. Support with safeguards.
13	B-6	SPEND(MAGALLANES) 980 to 833 · ModelB_B6_Peso	−1,558.2	48,687.8	−1,558.2	Yes	Yes (routing unchanged)	Robust to the peso; routing holds above about USD 550.
14	C-1 (A)	Add ALLOC(MAGALLANES, INFRA) ≥ 1.0 · ModelA_C1_MagInfra	−0.1324	121.3732	−0.1462	No	No (row dual 0.204)	Cost side of C-1: USD 0.146M.
15	C-1 (B)	HCAP(MAGALLANES) 10.6 to 11.6 · ModelB_C1_MagBeds1	+430.0	50,675.5	+429.5	No	Partly (to +0.9)	Benefit side: 1.29M over 2028 to 2030; 8.8 to 1.
16	C-1	Run 7 plus the forcing row of run 14 · ModelA_C1A7_Combined	n/a	122.9043	−0.0900 vs run 7	n/a	n/a (two changes)	With the office the move costs only 0.09M.
17	C-3	Binary PROMO, 0 or ≥ 1.0, at most 4 · ModelA_C3_Binary	n/a (integer)	121.0379	−0.4815	n/a	n/a	Atacama loses its campaign; no duals from a MILP.

Base objectives: Model A 121.5194 (USD million); Model B 50,246.0 (USD thousand). Every run is a separate LINGO file and Solution report in the zip, and each change one thing from its base model, except run 16, a deliberate two change check measured against run 7. A price from the model was exact only when the change stayed inside its Range report interval (runs 9 and 11) or the plan did not move (run 13). Supporting_Workbook.xlsx, tab Scenario_Log, recomputes every column with formulas.

Appendix E: Generative AI log

We used generative AI as a copilot throughout Phase 2, as the Terms of Reference encourage, in four chats: two in ChatGPT and two in Claude. This appendix lists each chat with its shared link or screenshots, logs every prompt with what the AI returned and what we did with it, and records the AI errors we caught. LINGO is the source of every number we report; nothing an AI computed outside LINGO was used without a LINGO run or our second solver behind it.

E.1 The four chats

Chat	Tool	What we used it for	Shared chat link or record
1	ChatGPT (course GPT)	Model B: line by line check of the LINGO code, scenario lines and what each dual price can and cannot predict, objective changes of the re-solves, SPEND and HCAP lineage.	chatgpt.com/share/e/6ac4a51e-9780-8327-83ba-be94139497d2
2	ChatGPT (course GPT)	Model A: the C-1 scenario code, the Appendix A formulation, and an independent cross check of the A-2 to A-5 and C-1 predictions against our re-solves.	chatgpt.com/share/e/6ac4a1eb-5150-8331-93fd-6d30ff826f0d
3	Claude	Model B verification from our LINGO files (cost orientation, lineage, dual prices and ranges, the B-2 reruns) and the C-3 binary extension.	Screenshots E1 and E2 (Section E.4)
4	Claude	Second solver check of every LINGO report and the A-1 comparison with Part 1.	Prompt log in Section E.2; 12 review passes recorded

E.2 Prompt log

#	Our prompt	What the AI returned	What we did with it
1.1	Verify the Model B code line by line: variable and row counts, five COST cells, inequality directions, the 13 OUTSIDE_CORE regions, the decentralization math, objective units.	Confirmed 90 variables, 37 constraints (38 rows), all five cells (gateways as rows), the four directions, 13 regions, 0.75 applied to all visitors, units USD thousand.	Matched against the LINGO Solution report (90 variables, 38 rows, global optimum 50,246.0).
1.2	Interpret the base Solution report (B-1).	The report did not paste; the AI asked for it instead of inventing values.	Answered B-1 from the LINGO report directly; checked in chat 3.
1.3	Interpret the Range report (B-2, B-3, B-7).	Same: no report received, no values invented.	Answered from the LINGO Range report; checked in chat 3.
1.4	Give the exact line to change for B-2, B-4, B-5 and B-6 and the prediction before solving.	Exact lines for each file. Flagged that B-5 changes a constraint coefficient and B-6 an objective coefficient, so an RHS dual price cannot price them directly.	Used those lines. In Appendix D, B-5 is marked n/a (coefficient) with an approximate prediction, and B-6 is priced as the coefficient change times the visitors routed.
1.5	Compare the re-solved objectives with the dual predictions.	Correct changes (+967.5, +1,781.0, −2,452.5, −92.5, −1,558.2) and noted the second 2,000 PUQ seats earn only 813.5. Would not judge ranges or routing without the reports.	Range calls and routing taken from the LINGO reports (Appendix D, rows 9 to 13).
1.6	Recompute SPEND and HCAP for Magallanes and Aysén from DB3.	SPEND 980 and 850 and Aysén HCAP 2.0 correct. Magallanes HCAP wrong: it computed 47,470 × 88 / 71.9 as 58,650.9 and reported a mismatch (11.2 against 10.6).	Rejected. The product is 58,099.6, so HCAP is 10.6 and matches Exhibit B2. Confirmed in chat 3 and for all 15 regions in the workbook (tab Lineage_DB3).
2.1	Write the full C-1 file: the base Model A plus [FORCE_MAGALLANES_INFRA], ALLOC (MAGALLANES, INFRA) ≥ 1.0, everything else unchanged.	A complete LINGO file with the base rules and the new row.	Run in LINGO as ModelA_C1_MagInfra: 121.3732, global optimum; equals the second solver. The same request was sent twice and returned the same file.
2.2	Write the Appendix A formulation of Model A with the exact row names; the floor is a flat 1.2, PROMO is relative to TALENT plus INFRA, macrozones come from our SETS block.	A full formulation: sets, 75 variables, parameters, objective and every constraint in words and math.	Checked against the code and condensed into the Appendix A table, keeping the flat 1.2 floor, the PROMO rule and the macrozone sets as written.
2.3	Independently cross check six predictions (A-2 up and down, A-3, A-4, A-5, C-1) and flag any wrong arithmetic or range call.	Predictions 137.3461, 105.6927, 117.8450, 120.8825, 119.4376 and 121.3870. A-2 and A-3 fall outside their ranges; A-4, A-5 and C-1 cannot be called within range. Reduced cost must be subtracted for C-1.	Adopted the range calls in Appendix D. Rejected its claim to have re-solved all six independently, which it cannot do without a solver; the objectives rest on LINGO and the second solver.
3.1	Verify the COST orientation on three cells, recompute SPEND and HCAP, rank the headroom duals and gateway seat values with ranges, explain the PUQ +4,000 result (files attached).	All checks passed; noted that PUQ to Magallanes reads 70 even in a transposed matrix; headroom ranking with ranges; PUQ worth 483.75 for 2,600 seats; 2.6 × 483.75 + 1.4 × 373.75 = 1,781.0.	Used for B-2 and B-3. Because five regions allow only 0.9 thousand more beds, B-3 is quoted per slot for 900 slots and Magallanes +1,000 was rerun in LINGO (+429.5 against 430).
3.2	Write the C-3 binary extension in LINGO and explain why its dual prices are not meaningful.	The binary model (90 variables, 15 integers, 103 rows), cost 0.4815, Atacama loses its campaign, and three reasons the duals mislead.	Same rows and optimum (121.0379) as our LINGO ModelA_C3_Binary report. Its Python figures for five campaigns (0.4797) and a forced Atacama campaign (0.3617) were left out because LINGO did not produce them.

#	Our prompt	What the AI returned	What we did with it
4.1	Complete the required files from the brief and the Data Pack; then read our Part 1 files.	A full draft package, a second solver and draft LINGO files, plus our RfP 2 split scored with Model A.	Treated as a draft only. The submitted LINGO models and reports are the team's own runs, and every number in this summary is read from those reports.
4.2	Check the three workstream.	Checked every LINGO report against the second solver.	Found 16 draft statements that disagreed with, or were not backed by, a LINGO report; all were checked against the reports.

E.3 What the AI got wrong, and how we caught it

Chat	What the AI produced	Why we did not accept it	What we did
1	Magallanes HCAP of 11.2 thousand and a claimed mismatch with Exhibit B2.	Arithmetic error: 47,470 × 88 / 71.9 is 58,099.6, not 58,650.9.	Recomputed by hand and in the workbook; HCAP 10.6 matches.
2	"All six re-solved objectives independently match your figures."	A chat model without a solver cannot re-solve an LP.	Relied on LINGO and the HiGHS second solver for every objective.
3	Costs of five campaigns (0.4797) and a forced Atacama campaign (0.3617) in C-3.	From its Python cross check, not LINGO.	Not reported; only the LINGO C-3 result is quoted.
3	Value of 1,000 extra beds read straight from the headroom duals.	Five regions allow only 0.9 thousand before the dual changes.	Quoted per slot for 900 slots; reran Magallanes +1,000 in LINGO.
4	Briefing line that both models "match the Ministry's reference answers".	No file gives reference values.	Replaced with "solved to a proven optimum in LINGO".
4	Equity floor money described as earmarked to training, as a model result.	The LINGO run has no earmark; USD 0.21 million goes to PROMO in Metropolitana.	The earmark is presented as our policy recommendation.
4	Single change results shown together as if they held jointly.	Run together, the floor and the Magallanes move give a different split.	Each result is stated under today's rules, one change at a time.
4	A-1 text: Metropolitana's multipliers are "the lowest in Chile".	True only of INFRA and NATURE.	Corrected in Section 1.

E.4 Screenshots of chat 3 (Claude)



E1. Model B check: cost orientation, lineage, dual prices and the B-2 reruns.



E2. C-3: the binary PROMO extension and why its dual prices are not meaningful.

What the AI got right, what we corrected, and how we verified it. The AI wrote working LINGO code with SETS, DATA and named constraint families, confirmed our cost matrix orientation cell by cell, correctly warned that dual prices cannot price the B-5 and B-6 changes directly, and reconciled the Punta Arenas reruns exactly (2.6 × 483.75 + 1.4 × 373.75 = 1,781.0). It also made real errors: a wrong Magallanes headroom of 11.2 thousand, a claim to have re-solved models it could not solve, C-3 figures that came from Python rather than LINGO, and briefing claims with no file behind them, all of which we rejected or corrected (Section E.3). Every number in this summary was read from the LINGO Solution and Range reports, and an independent second solver (HiGHS, in the zip) reproduced all 546 numbers of the two base reports and the objective of every scenario. Each scenario was re-solved in LINGO

instead of extrapolating a dual price beyond its range. The interpretation, the recommendations and the story are our own judgment.